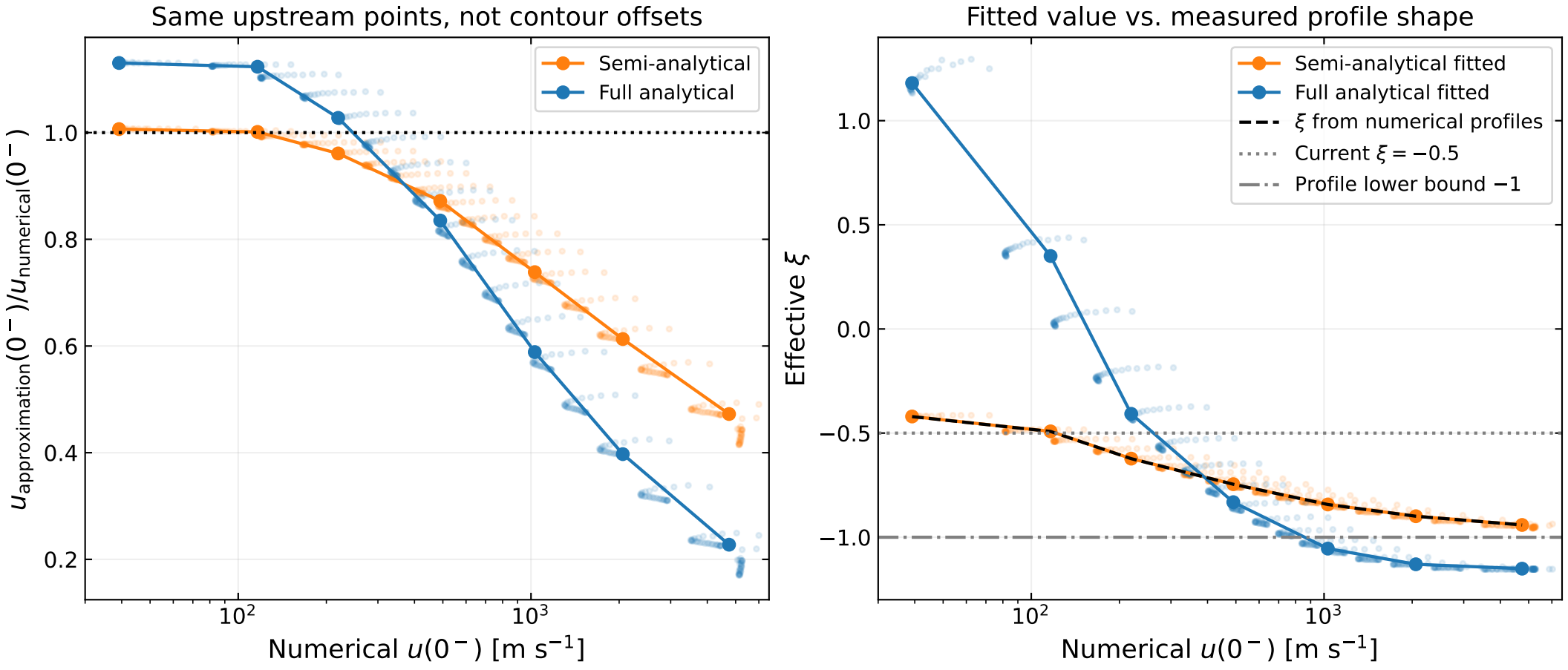

Same upstream points, not contour offsets
Semi-analytical
Full analytical
$u_{approximation}(0^-)/u_{numerical}(0^-)$
1.0
0.8
0.6
0.4
0.2
10^2
10^3
Numerical $u(0^-)$ [m s^{-1}]
Fitted value vs. measured profile shape
Semi-analytical fitted
Full analytical fitted
ξ from numerical profiles
Current $\xi = -0.5$
Profile lower bound -1
Effective ξ
1.0
0.5
0.0
−0.5
−1.0
10^2
10^3
Numerical $u(0^-)$ [m s^{-1}]